

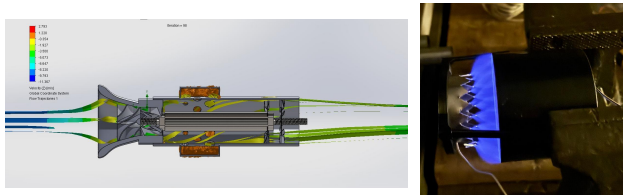
Ion the Prize: Turbojet Efficiency

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Scientific Question

One of the greatest challenges currently confronting our society is the transition from fossil fuels. I have created this project in hopes to facilitate this shift by examining the effects of ionized air intake into a model turbojet engine. The study seeks to analyze how this process affects the engine’s operational parameters, overall performance and longevity.

Figure 1 (Left) - Flow simulation
Figure 2 (Right) - Corona discharge plasma from ion thruster



Methodology

- Organization:** Fuels were separated into six cleaned, and independent gas cans (Figure 1). The carburetor bowl of the generator was emptied each trial to ensure no fuel was remaining in the tank (Figure 2).
- Measuring equipment:** Measuring equipment consisted of a model turbojet engine, thrust measuring apparatus and an arduino UNO with supporting interfaces. The turbojet was mounted on a bearing pivot arm with a 3:1 mechanical advantage against a 1kg load cell. The load cell was able to measure to an accuracy of 0.01(g) which gave me a thrust measurement. There was also a thermocouple placed on the inside of the engine, behind the combustion chamber, allowing internal temperatures to be monitored. Further, a gas sensor was used to quantify the amount of harmful emissions put out by the combustion process. Finally, a current transducer was added to measure the power draw of the ion thruster when applicable. All of these measure data points were recorded by the arduino every 500ms, and output to a serial monitor on a laptop where they could later be compiled and organized into a spreadsheet.
- Data:** Data was collected using the measuring equipment mentioned. About 2,400 data points in total were collected, organized, and visualized using spreadsheet software. Some data was also extracted using multiple data points such as total power production, draw or rate of temperature increase. 24 unique graphs were created in total.
- Comparison:** Data was compared in several different ways. Total power output over all 90 seconds of each test series. This allowed me to compare total power generation. Also, average gas sensor reading was used to quantify the difference in combustion efficiency. Moreso, temperature and rate of temperature increase were compared to analyze operation.

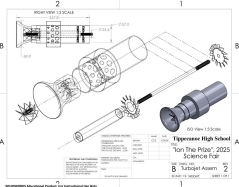
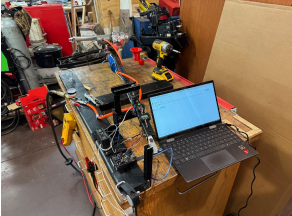


Figure 7 (left) - Draft of exploded view of turbojet.

Figure 8 (right) - Testing setup, from load cell end.



Data Analysis & Results

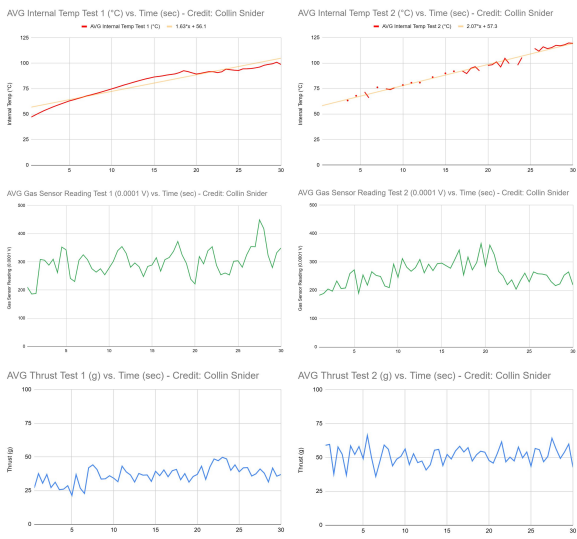


Figure 3 (left)
Test 1 AVG temperature vs time over 3 trials.

Figure 4 (right)
Test 2 AVG temperature vs time over 3 trials.

Figure 5 (left)
Test 1 AVG gas sensor reading vs time over 3 trials.

Figure 6 (right)
Test 2 AVG gas sensor reading vs time over 3 trials.

Figure 7 (left)
Test 1 AVG thrust (g) vs time over 3 trials.

Figure 8 (right)
Test 2 AVG thrust (g) vs time over 3 trials.

Interpretation & Conclusion

- The use of ionizing components on the intake stage may be beneficial for the power output and reduction of harmful emissions, whilst improving power output with the drawback of higher internal temperatures.
- As shown in the graphs above, the average internal temperature of the tests of which the intake air was ionized showed to increase at a greater rate and reach a higher maximum compared to the test which did not involve intake ionization.
- In addition, the average reading from the gas sensor in the test with ionized air produce values ~13.8% less than that of normal atmosphere. It should be noted these data points have a deviation of -2% or +4% across all trials, showing great confidence.
- Finally, most significantly, the thrust output by the turbojet engine increased by a factor of 40.84% on average in the period of the six trials. This also translates to a 40.84% total power output increase.
- Further, upon final tear down of the engine, there were no signs of advanced corrosion by the ionizing air, giving promise that the process doesn’t have much if any effect on longevity of the engine.
- Overall, these results proved that by ionizing the air to the intake stage of a turbojet engine could result in higher power, less harmful emissions, with a moderate internal temperature increase. This shows promise that this process may be beneficial in solving our climate crisis and potentially paving way to more efficient air travel and engine operation in general.